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Structured Runtime Safety Monitoring for Camera-Based Driver Monitoring Systems

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***Abstract*—This paper presents a structured runtime safety monitor for camera-based driver monitoring systems (DMS) in automotive and other safety-critical domains. The proposed approach enhances AI safety by combining real-time perception with layered semantic reasoning to improve explainability and fault detection, consistent with ISO/PAS 8800[1], ISO 26262[2], and ISO 21448[3] requirements. A hybrid L1–L2 architecture fuses a lightweight perception layer (Level 1) with Chain-of-Thought (CoT) based semantic reasoning (Level 2) to increase reliability and interpretability under real-time constraints. Experiments on the YawDD dataset show that the hybrid architecture achieves 92.16% accuracy and significantly reduces false alarms compared to perception only baselines, without violating end-to-end timing constraints.**

***Index Terms*—Artificial Intelligence Safety, Structured reasoning, Functional Safety, ISO 26262, ISO 8800, Explainable AI (XAI), SOTIF, ISO 21448**

I. Introduction

AI models such as machine learning (ML) lack safety guarantees, explainability, and predictable failure modes, making them challenging for safety-critical applications. The proposed structured reasoning based monitoring approach helps to achieve functional safety for AI-driven safety critical systems. This method tackles key safety concerns such as traceability, fault detection, and justification of AI outputs which aligns with the AI explainability requirements from ISO 8800[1]. Under ISO 26262[2], safety-related functions must demonstrate hazard mitigation, timing guarantees, and traceable decision logic. ISO 21448 (SOTIF)[3] addresses performance limitations and unintended behavior arising from insufficient situational understanding. Pure deep learning based perception systems struggle to satisfy the above standards due to semantic ambiguity and opaque reasoning.

Driver Monitoring Systems (DMS) are essential for safe vehicle takeover in ADAS-assisted driving and typically require at least ASIL B compliance. The geometric feature–based DMS implementations (e.g., InsightFace[4]) achieve high accuracy in controlled settings, but their deep learning paradigms suffer from a “semantic gap” complicating safety validation. Geometric-based DMS faces two key limitations such as (1) Ambiguity of physical representation is when there is difficulty distinguishing similar actions such as "yawning" and "talking", leading to high-frequency false alarms.(2) Inability to detect implicit fatigue e.g.,cognitive lapses or unconscious driving[5].

Recent Visual-Language Models (VLMs) with Chain-of-Thought reasoning[6, 7, 8] can address the semantic gap but suffer from excessive inference latency (on the order of seconds), making them unsuitable for real-time applications. To cope with limited computational resources, several hybrid frameworks have adopted a serial switching paradigm in which a fast perception system triggers a slower semantic reasoning system. Examples include the FASIONAD framework[9], which activates a VLM when trajectory prediction confidence is low, and Anomaly detection systems[10] that invoke large models through out-of-distribution triggering. Current DMS designs[11] also use geometric perception as a wake-up mechanism for multimodal large models.However, such serial architectures are susceptible to cascading failures e.g., when the perception layer misclassifies a fatigued but visually alert driver as “safe”, the higher level reasoning module remains inactive, leading to false negatives. Moreover, the cold-start delay of large models hinders timely response in time-critical scenarios. To overcome these limitations, this paper proposes an asynchronous concurrent logic architecture that maintains full-time parallelism between rapid geometric feature extraction and continuous semantic feedback from a VLM. This design ensures that the high-level semantic baseline remains pre-initialized, effectively eliminating reliance on physical thresholds and addressing hidden fatigue false negatives as well as cold-start latency.

Contributions — This paper proposes a deterministic,

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multi-step runtime safety monitor for camera-based DMS, featuring:

- A hybrid real-time and semantic reasoning DMS architecture aligned to functional safety[2], SOTIF[3] and AI safety[1].
- Quantitative evaluation on the YawDD dataset.
- A structured framework for integrating explainable reasoning into automotive AI safety systems.

II. BACKGROUND

Deep learning based DMS methods[12, 13, 14]typically use CNN or Transformer architectures for gaze estimation, eye-closure detection, and head-pose tracking. Although these methods offer high frame-level accuracy, they often lack temporal awareness and contextual reasoning. VLMs[15] and other multimodal systems have recently demonstrated strong reasoning ability, yet their integration into real-time safety-critical systems remains limited.

ISO 26262 mandates safety mechanisms, while ISO/PAS 8800 addresses AI-specific concerns such as data bias, epistemic uncertainty[16], and lack of explainability[17]. ISO 21448 (SOTIF)[3] further focuses on risks from inadequate perception and understanding.

This work bridges low-level perception with structured semantic reasoning to achieve ASIL-B compliance. It introduces an asynchronous concurrent monitoring architecture that decouples perception and cognition pathways, prevents cascading failures, achieves global semantic feedback, and ensures freedom from interference. The targeted safety requirement is *"The driver monitoring system shall detect and escalate driver inattention within 3.0 seconds under all single-point fault conditions and foreseeable performance limitations.-ASIL B"*.

III. METHODOLOGY AND SYSTEM ARCHITECTURE

To resolve the contradiction between the lack of semantics and the fatal cold-start delay and single-point false negatives commonly observed in traditional systems, this paper proposes a continuously operating asynchronous concurrent architecture, as illustrated in Fig. 1. The system consistes of two independent, parallel models- L1(Perception) and L2(semantic reasoning)and a global finite-state machine fusion module.L1 and L2 exhibit distinct failure modes, and their integration forms a heterogeneous redundancy that enhances overall system reliability. When L1 fails, L2 can continue the decision process, and conversely, when L2 fails, L1 remains functional to maintain continuous operation. In order to achieve freedom from interference,L1 and L2 are deployed on separate processing cores within a multi-core SoC designed for ASIL B applications. L2 and fusion module is deployed in safety core and is ASIL B compliant.The L1 is not developed as per functional safety and is hence QM(B) to achieve ASIL decomposition of ASIL B.The L2 and fusion modules are supervised by a program health monitoring mechanism that performs alive monitoring, logical monitoring, and deadline monitoring.L2 performs cross-checking of L1 outputs while remaining architecturally independent.As DMS is critical for ADAS operation in supervised mode,the vehicle must exit the ADAS function when an error is reported from health monitoring. The safety design for DMS is as follows:

- **L1 Perception Layer QM(B):** Operates continuously in real-time using lightweight neural network algorithms for facial key-point extraction, providing ultra-low-latency risk monitoring. L1 runs at 30 FPS to deliver instantaneous preliminary assessments with minimal computational overhead.
- **L2 Semantic Reasoning Layer -ASIL B(B):** Executes asynchronously in a background thread at a lower frequency, performing continuous high-level inference through a VLM. The inputs include L1 classification outputs, L1 confidence scores, and temporal frame sequences. L2 reasoning tasks comprise temporal consistency checking, eye-closure duration analysis, head-pose stability evaluation, and cross-frame contextual reasoning. The layer produces semantic outputs such as *confirm alert*, *false-alarm suppression*, or *escalation to driver warning*.
- **Conservative Strategy Decision Fusion -ASIL B:** Integrates historical outputs from both L1 and L2, incorporating semantic cues derived from sequential video data to compute the real-time confidence level of L1 and determine the final safety action.

A. L1 Perception Layer

The L1 layer follows a minimalist, ultra-fast design aimed at instantaneous driver state recognition. Based on real-time video input, it classifies the driver state into three discrete categories *normal*, *L1 fatigue*, and *L1 drowsiness*. The framework adopts *InsightFace* as its base model for face detection and 2D landmark estimation. To guarantee computational efficiency, frames are processed at a constant inference rate, yielding a 106-point facial mesh. Using these landmarks, two primary physiological indicators are calculated:

- EAR (Eye Aspect Ratio): Quantifies eye opening to detect closure and momentary drowsiness.

$$\mathrm{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|} \tag{1}$$

  Where: p_1, p_4 are the left and right inner/outer canthus points of the eye (horizontal width); p_2, p_3, p_5, p_6 are the upper and lower eyelid feature points (vertical height).
- MAR (Mouth Aspect Ratio): Measures the mouth's aspect ratio to capture yawning events

$$\mathrm{MAR} = \frac{\|p_2 - p_8\| + \|p_3 - p_7\| + \|p_4 - p_6\|}{3\|p_1 - p_5\|} \tag{2}$$

  Where: p_1, p_5 are the left and right mouth corner points (horizontal width); p_2, p_3, p_4 and p_6, p_7, p_8 are the vertical feature points on the upper and lower lip, respectively.

To mitigate high-frequency noise introduced by vehicle vibration or natural blinking, L1 employs a multi-scale temporal smoothing and aggregation strategy for robust state estimation.

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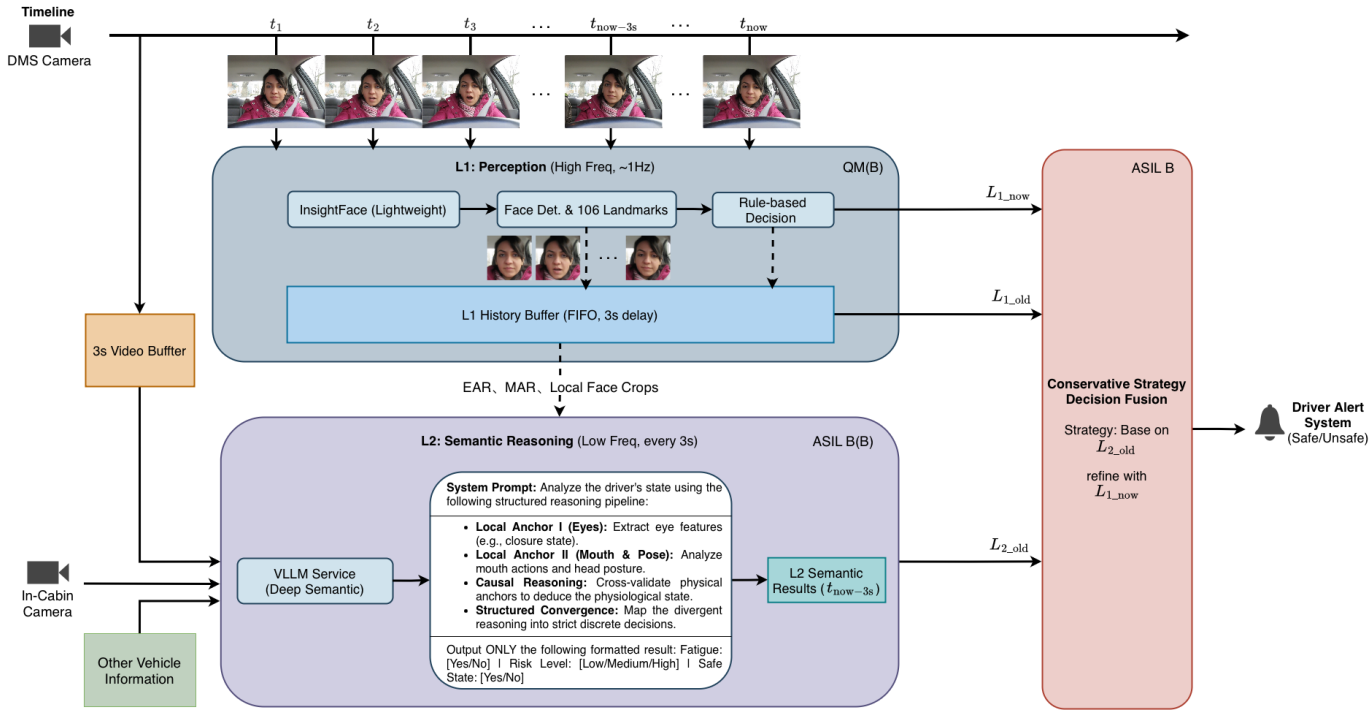

Fig. 1. Proposed DMS safety architecture

Internally, a sliding-window filter is applied over fixed time steps (*Tick*). If the median EAR within the window is below 0.2 or if the proportion of low-EAR samples is ≥ 0.5, the Tick is labeled as a drowsiness candidate. Similarly, if MAR remains above a dynamic threshold for five consecutive frames, a yawning event is registered. Repeated yawns occurring within a short interval are merged into a single fatigue event.

When eye occlusion occurs (e.g., sunglasses), EAR-based evaluation is adaptively disabled and inference relies on yawning-related features. To enhance stability, hierarchical temporal aggregation is introduced is applied: if $\geq 75\%$ of Ticks within a window indicate drowsiness, an "L1 Drowsiness" state is triggered. If the current frame lies within a previously triggered "Fatigue Window, " the state is determined as "L1 Fatigue." L1 aggregates results every 3 seconds,using a strict priority rule of *Drowsiness* > *Fatigue* > *Normal*, thereby ensuring stable and interpretable input to downstream safety-critical modules.The L1 module is developed as per Quality Management QM(B) without functional safety requirements.

B. L2 Semantic Reasoning Layer

The core objective of the L2 Asynchronous Semantic Understanding Layer is to leverage the VLM for deep semantic scene understanding. To mitigate the common issues of hallucination and zero-shot reasoning uncertainty exhibited by VLMs when dealing with global field-of-view inputs, this paper introduces a multi-granularity visual–textual prompting inference framework.

At the visual representation level, the L2 layer encodes both global context and localized features by representing its input as temporally aligned Global Context Frames and Local Face Crops. Furthermore, to overcome the modal limitations of pure visual reasoning, deterministic numerical features from L1 such as Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head-pose angle are converted into textual priors and dynamically injected into the VLM's prompt space. These rule-based indicators act as numerical anchors that stabilize the VLM's visual perception, effectively suppressing divergent reasoning and hallucinations under challenging conditions such as low illumination or partial occlusion fulfilling SOTIF requirements.

At the cognitive reasoning level, to address the "sycophancy" problem in end-to-end generative inference typical of large models, a Structured Chain-of-Thought (S-CoT) mechanism is proposed. Through customized prompt engineering, this mechanism enforces four explicit stages of cross-modal, cross-validation reasoning before generating a final decision: **Local Anchoring I (Eye Features):** The VLM combines facial cropped frames with the EAR metric at the L1 to finely evaluate the eyelid closure state.
Local Anchoring II (Mouth and Head Pose): The model compares MAR metrics with visual sequences to determine mouth micro-movements such as yawning or talking, and evaluate the driver's attention focus based on head pose angles;
Causal Inference (Cross-Validation): This step comprehensively integrates the above multi-granularity visual cues and physical values for logical deduction.It can accurately distinguish easily-confused corner cases such as "singing loudly (high MAR but with a clear state of mind)" and "yawning due to severe fatigue (high MAR accompanied by low EAR)".

TABLE I
ACTION LOOKUP TABLE

$L2_{old}$	$L1_{old}$	$L1_{now}$	action
1	1	1	1
1	1	0	0
1	0	1	0.5
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	1
0	0	0	0

Structured Convergence: Aggregates diverse reasoning outcomes into discrete, interpretable decisions (e.g., *Fatigue Status: Yes/No*, *Risk Level: Low/Medium/High*, *Safety Status: Yes/No*).

Although L2 incorporates numerical anchors from L1 (e.g., EAR, MAR, and pose), these signals are treated as auxiliary context rather than authoritative inputs.Common cause failures are eliminated by this approach. The L2 decision logic primarily relies on raw sensory evidence from the driver-facing camera, optionally augmented with additional in-cabin views to capture a broader context (body posture, hand activity) and supplementary vehicle-state signals (e.g., steering torque, vehicle speed, and ADAS engagement). This multi-modal integration enhances functional diversity and minimizes dependency on any single perception channel.

The hierarchical reasoning from local anchoring to global semantic inference not only provides vehicle-grade explainability (XAI) but also significantly enhances robustness under complex operational scenarios. By introducing an asynchronous semantic monitor, the proposed architecture narrows the Area of Unknown Unsafe Scenarios (Area 3 in ISO 21448). When the L1 perception model reaches performance limits such as ambiguous expressions or heavy occlusion, L2 applies contextual reasoning to preserve the intended safety functionality.The L2 modue is implemented as per ASIL B.

C. Conservative Safety Decision Fusion

The Fusion module of this system abandons the traditional blocking wait and instead constructs a lightweight finite-state machine based on $(L2_{\text{old}}, L1_{\text{old}}, L1_{\text{now}})$ time-series triples. During the decision-making phase, the system calls the historical semantic state of the L2 layer in combination with the historical information and current output of L1 to achieve the current response. To balance the driving safety baseline and the human-machine interaction experience, this paper designs an asymmetric arbitration truth table (as shown in Table I) that includes "Soft Warning".The fusion module is implemented as per ASIL B.

Here, state 1 denotes “safe, ” 0 denotes “danger” (triggering a high-level alarm), and 0.5 corresponds to a soft or early warning. The core FSM logic is governed by four key mechanisms:

- **L1 Veto Logic:** When $L1_{\text{now}} = 0$ (Table I rows 2, 4, 6, 8), the system immediately outputs 0 regardless of history. This ensures rapid intervention when explicit physical features such as eye closure or head drop are detected.
- **L2 Semantic Coverage Mechanism:** Corresponds to the state where $(L2_{\text{old}} = 0, L1_{\text{old}} = 1, L1_{\text{now}} = 1)$ outputs 0. When L1 determines the physical state to be safe for two consecutive frames, but L2’s semantic inference indicates danger, the system still outputs an alarm. This allows L2’s temporal reasoning to overrule L1 in edge cases like “micro-sleep with open eyes” or partial occlusion, addressing L1’s rule-based limitations.
- **State Debounce and Flexible Early Warning:** Corresponds to the state with an output of 0.5 for $(L2_{\text{old}} = 1, L1_{\text{old}} = 0, L1_{\text{now}} = 1)$. When the physical characteristics suddenly change from dangerous to safe and the L2 historical semantic base line determines it to be safe, the system attributes this fluctuation to sensor noise or driver transient actions (such as rapid blinking) and outputs a flexible state of 0.5 to filter out high-frequency false alarms.
- **State Quick Reset:** Corresponds to the state where $(L2_{\text{old}} = 0, L1_{\text{old}} = 0, L1_{\text{now}} = 1)$ outputs 1. When both the historical states of L1 and L2 are dangerous, indicating that L1 is operating normally and the physical characteristics of the current frame have clearly returned to normal, the system directly resets the output to 1.

In summary, the ternary finite-state fusion mechanism achieves deterministic, low-latency decision arbitration with minimal computational overhead. This design minimizes false alerts while ensuring timely intervention and maintaining compliance with automotive functional safety requirements.

IV. EXPERIMENTS

A. Dataset

Experimental validation was conducted using the YawDD dataset. The raw videos were segmented into 3-second temporal windows sampled at 4 fps, yielding 12 frames per segment. As summarized in Table II, the test set contains 354 segments (totaling approximately 1062 s). Different from many settings that only report overall accuracy, this paper focuses more on the performance in hard and rare examples. The reason is that methods based on rules or lightweight detectors often already achieve high accuracy on "easy examples", while the risks of actual systems often come from low-frequency scenarios such as occlusion, glasses reflection, and performative actions. The introduction of L2 is precisely to provide supplementary semantic-level judgments in situations where the boundaries of these rules are unstable. To adapt the dataset to the semantic reasoning task of L2, we developed a GPT-4o-driven semantic annotation pipeline with human-in-the-loop verification. This ensured that segment-level labels accurately reflected real semantic intent rather than purely geometric cues.

TABLE II
DATASET DISTRIBUTION BY GENDER AND GLASSES

Category	Segments	Dur. (s)	Prop.
Female / No Glasses	179	537	50.56%
Male / Wearing Glasses	175	525	49.44%
Total	**354**	**1062**	**100%**

B. Setting

All experiments were conducted on an x86 cloud platform equipped with an NVIDIA RTX 3090 (24 GB), implemented in PyTorch using dynamic computation graphs. The L2 VLM module (Qwen3-VL-8B-Instruct) employed BF16 precision during inference.

In contrast, in-vehicle deployment typically targets edge platforms such as Jetson Orin or Jetson Thor, utilizing static-graph execution via ONNX and TensorRT, together with low-precision quantization (e.g., INT8/FP8/INT4). Consequently, the runtime characteristics differ from the cloud baseline.

To assess transferability, we apply scaled argumentation. The RTX 3090 provides peak BF16 tensor throughput of 71/142 TFLOPS (dense/sparse) with 936 GB/s memory bandwidth, whereas Jetson Thor delivers up to 2070 TFLOPS (FP4-sparse) AI inference performance and 273 GB/s memory bandwidth[18]. Although the RTX 3090 offers higher memory bandwidth, this primarily benefits long-sequence token decoding. In this work, the L2 module performs fragment-level asynchronous review with short outputs, making it more sensitive to compute efficiency than memory bandwidth.

Therefore, if latency constraints are satisfied under the cloud configuration (3090 + BF16 + dynamic graph), optimized in-vehicle deployment using static graphs and low-precision quantization is expected to achieve higher throughput and lower latency, while maintaining stable functional performance after full INT8 calibration.

C. Results and Discussion

Table III presents the benchmark comparison between our proposed hybrid framework (L1+L2) and the standalone L1 (rule-based) and L2 (VLM-based) modules.

The results show that the L1 module provides millisecond-level fast response (0.075 s average inference time), but is limited in detection effectiveness (Accuracy = 0.7922, Recall=0.6667); after introducing VLM, the L2 module significantly improves its performance (Accuracy = 0.9118, Recall = 0.8000), indicating that segment-level semantic modeling can effectively compensate for the insufficient expression of rule-based methods in complex behavior scenarios; On this basis, the L1+L2 hybrid architecture further boosts Accuracy to 0.9216, indicating that the double-layer asynchronous fusion framework proposed can further enhance overall detection performance without introducing additional L2 latency overhead. This improvement does not simply come from adding a stronger model, but from changes in the information organization method and decision-making mechanism. This can be explained from several aspects: Face cropping, as an explicit attention prior, improves the signal-to-noise ratio of facial cues, enabling the VLM-based L2 module to focus more on task-relevant regions. Additionally, the history buffer temporally aligns L1old and L2old, allowing L2 to serve as a posterior semantic validator rather than a delayed substitute for L1. The conservative fusion strategy further stabilizes the final decision by combining the real-time responsiveness of L1 and the delayed but semantically richer evidence of L2. The slight drop in recall (0.75 vs 0.8) is a deliberate trade-off of our Conservative Fusion Strategy, prioritizing the prevention of high-frequency false alarms over extremely high sensitivity.

Rule-based methods have the advantages of low latency and easy deployment, but their judgments often rely on geometric quantities and fixed thresholds. In hard examples, these factors can change the image projection of local features, causing behaviors that are "shape-similar but semantically different" to be misjudged as fatigue states. While supervised learning methods based on Deep learning have reported high accuracy on specific datasets, these methods are essentially limited by the training distribution of the closed set. In contrast, the purpose of introducing L2 in this paper is not simply to replace L1, but to perform posterior review of L1 through segment-level spatio-temporal semantic analysis. The VLM utilized by this framework possesses pre-trained world knowledge and zero-shot reasoning capabilities. Even if it has not encountered a specific scenario during the training phase, the VLM can still adapt naturally to long-tail scenarios through reasoning.

In error analysis, we discovered a counterintuitive phenomenon, which is also the main reason for Recall being slightly lower than expected: VLM misclassified some samples with ground truth labeled as "fatigue" as "awake". After manual review, these samples clearly showed signs of severe acting. Existing fatigue datasets are basically recorded by awake subjects imitating fatigued actions. Although subjects can imitate "yawning", they often struggle to reproduce the physiological micro-features associated with real fatigue (such as distracted eyes, relaxed eye muscles, tearing, etc.), and may even unconsciously smile during the acting process. Traditional models are prone to overfitting these "performance features", thus achieving inflated accuracy on these datasets. In contrast, the VLM model, pre-trained on massive real human behavior data, keenly captures the semantic conflict between "performance" and "reality", resulting in seemingly lower accuracy. Therefore, we believe that the seemingly lower recall of the L2 module is actually a correction of noisy ground truth. The measured inference latencies show that L1 detection completes within 75 ms, while L2 semantic reasoning operates over a 1.85 s asynchronous window, resulting in a maximum end-to-end latency of 2.28 s. This total response time lies well within the human reaction window of 2–3 s and conforms to the ASIL B FTTI (Fault Tolerant Time Interval) requirement.

TABLE III
PERFORMANCE AND LATENCY COMPARISON

Experiment	Scene Description	Accuracy	Recall(unsafe)	Avg. Inference(s)	Max Inference(s)
L1+L2	Face as noticed feature	0.9216	0.75	1.85	2.28
L2_only	L2 individual summary	0.9118	0.8	1.85	2.28
L1_only	Baseline/Comparison group	0.7922	0.6667	0.075	0.092

V. CONCLUSION AND FUTURE WORK

This paper proposes a continuously operating asynchronous L1–L2 architecture that improves Driver Monitoring System functional safety by avoiding missed latent-fatigue detections and early-warning delays while remaining consistent with ISO 26262 requirements. Experiments on the YawDD dataset show that the hybrid design achieves 0.9216 accuracy, and its Structured Chain-of-Thought mechanism anchors VLM reasoning to EAR/MAR evidence, strengthening AI safety and explainability in line with ISO/PAS 8800 and XAI goals. By adding an asynchronous semantic monitor, the framework also reduces the Area of Unknown Unsafe Scenarios defined in ISO 21448, enhancing SOTIF performance when L1 reaches its limits. The architecture follows an "architecture-as-a-service" principle that decouples safety logic from specific model instances, allowing future vision-language model upgrades without redesigning the safety concept. Future work will focus on pruning, quantization, and acceleration to deploy this ISO 26262 and ISO 21448 compliant AI safety monitor efficiently on in-vehicle hardware.

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